Doublers are riveted to the forward and aft sides of the spar at the attachment location and outboard beyond the bend in the spar that forms the wing dihedral. Flanges for the doublers extend over the upper and lower spar caps at the forward side of the spar and over the lower spar cap at the aft side of the spar.

For reference in this report, wing spar attachment bolts were labeled with three alphanumeric characters to identify their locations. The first letter in the identification was an L or an R, indicating left or right wing, respectively. The second letter in the identification indicated the row of bolts consistent with the labeling method used in the Piper Aircraft, Inc. PA-28R-201 Arrow Maintenance Manual. Attachment locations for the upper spar cap were identified with A and B, where row A was located forward of the spar web and row B was located aft of the web. Attachment locations for the lower spar cap were identified with C and D, where row C was located forward of the spar web and row D was located aft of the web. Finally, the bolts in each row were identified with a number in sequence starting from the outboard bolt, also consistent with the labeling method used in the aircraft maintenance manual. As in the manual, a dash is inserted between the row letter and the number. As an example, the outboard attachment bolt in the lower spar cap forward of the web in the left wing would be labeled LC-1.

Views of the fracture in the left spar are shown in figures 2 and 3. The lower spar cap shown in figures 2 and 3 had a fracture that intersected holes for attachment bolts LC-1 and LD-1. Portions of the fracture surface had relatively smooth features in a plane perpendicular to the longitudinal axis, features consistent with fatigue fracture. Crack arrest lines were visible on the fracture surfaces of the lower spar cap and the forward and aft doublers, also consistent with fatigue fracture. Some of the fracture features were obscured by black-colored deposits, particularly in the areas near the apparent fatigue origins. The upper spar cap displayed rough matte-gray features consistent with ductile overstress fracture. Deformation associated with the fracture in the upper spar cap was consistent with an upward bending load.

The outboard side of the fracture shown in figure 3 was cut from the rest of the spar to facilitate further examination. Additionally, the forward and aft doublers were removed from the sectioned piece by drilling out several rivets. To remove the black deposits from the surface, acetate replica tape was applied to the surface and removed after drying. After several applications of the replica tape, the pieces were placed in a bath of acetone and cleaned using an ultrasonic cleaner. A view of the lower spar cap and doubler pieces after sectioning, separating, and cleaning is shown in figure 4 with closer views in figure 5.

Unlabeled arrows in figure 5 indicate locations of fatigue origins, and dashed lines indicate fatigue boundaries. In the lower spar cap, origins were located at the lower surface both forward and aft of hole LC-1 and at the aft side of hole LD-1. Forward of hole LC-1, fatigue features extended to the forward edge of the lower spar cap as shown in a closer view of the fracture surface in figure 6. Aft of hole LC-1, fatigue features extended aft through the spar cap, intersecting the lower side of hole LD-1, and upward into the web to the boundary located approximately 0.75 inch above the spar cap. From